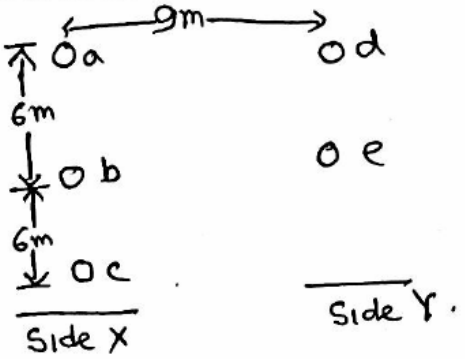
Power System II

TT-1

(a) Between aluminum and copper which one is superior? Explain the reason behind it?

(b) One circuit of a single-phase transmission line is composed of three solid 0.25 cm radius wires. The return circuit is composed of two 0.5 cm radius wires. The arrangement of conductors is shown in Fig.1. Find the inductance due to the current in each side of the line and the inductance of the complete line in henrys per meter.



Exp 4.2
Stevenson
page 158

Fig: 1

(c) Find the capacitive susceptance per mile of a single-phase line operating at 60 Hz. The conductor is Partridge, and spacing is 20 ft between centers.

Power System II

TT-2

1. Discuss the desirable properties of insulators
2. What is String efficiency? Discuss the methods of improving string efficiency
3. Classify the cables depending on the voltage capacity
4. An 11 kV, 50 Hz, single phase cable 2.5 km long, has a diameter of 20 mm and internal sheath radius of 15 mm. If the dielectric has a relative permittivity of 2.4, determine (i) capacitance (ii) charging current(iii) total charging kVAR.

11.3 - 277 page - Mehta

Shahjalal University of Science and Technology
Department of Electrical and Electronic Engineering
3rd year 2nd Semester Final Examination December 2021

Course Code: EEE 337
Course Title: Power System II

Time :3hours
Total Marks : 100

Answer any two questions from each part

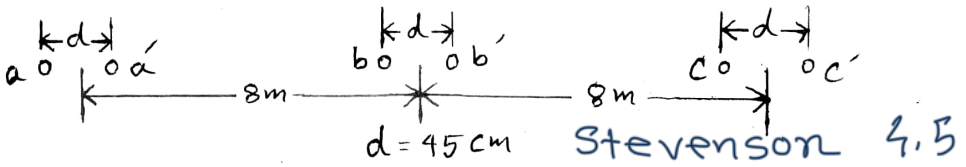
Part A

1. a) Prove that the inductance of a single-phase two-wire line is $L=4\times10^{-7}$ ln D/r´ H/m 5

   b) Which one is superior between aluminum and copper conductor and also explain why? 5

   c) What is the difference between composite conductor and bundled conductor? 5

   d) Each conductor of the bundled –conductor line shown in Fig.1 is ACSR, 1272000-cmi Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. Also find the per-unit series reactance of the line if it's length is 160 Km and base is 100 MVA, 345 KV ,f=60 Hz 10



Stevenson 4.5

Fig.1

2. a) What is transposition and why transposition is used in transmission line? 2

   b) Show that in a single core cable $g_{max}/ g_{min}=D/d$, where the symbols have their usual meaning. 10

   c) Calculate the capacitance to neutral in farads per meter of a single-phase line composed of two solid circular conductors each having a diameter of 0.229 in. The conductors are 10ft apart and 25ft above ground 3

   d) Calculate the capacitance and charging current of a single core cable used on a 3- phase, 66 KV system. The cable is 1 Km long having a core diameter of 10 cm and an impregnated paper insulation of thickness 7 cm. The relative permittivity of the insulation may be taken as 4 and the supply at 50 Hz 10

Exp 11.5

3. a) Determine the insulation resistance of a single core cable of length 3 Km and having conductor radius 12.5 mm, insulation thickness 10 mm and specific resistance of insulation of 5×10^{12} Ωm 8

Tutorial 11.3

   b) Find out the most economical conductor size in a cable 7

   c) A 33 KV, 50 Hz, 3-phase underground cable, 4 km long uses three single core cables. Each of the conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5 cm. Determine (i) capacitance of the cable/phase (ii) charging current/phase (iii) total charging KVAR. The relative permittivity of insulation is 3 10

Exp 11.6

Part B

4. a) What is String efficiency? Discuss the methods of improving string efficiency 10

   b) Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 KV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency. 10

Exp 8.3

   c) Discuss the causes of insulation failure 5

5. a) What is Corona? Explain the theory of Corona formation. Discuss the method of reducing of Corona effect 8

b) Consider a 132 KV line with 1.956 cm diameter. The conductors are built so that corona takes place if the line voltage exceeds 210 KV (rms). If the value of potential gradient at which ionization occurs can be taken as 30 KV per cm, find the spacing between the conductor 7

8.14

c) An overhead transmission line at a river crossing is supported from two towers at heights of 40m and 90m above water level, the horizontal distance between the towers being 400m. If the maximum allowable tension is 2000 kg. find the clearance between the conductor and water at a point mid-way between the towers. Weight of conductor is 1 Kg/m 10

8.25

6. a) Mention the types of HVDC system 5

b) Derive the swing equation 10

c) A rectangular pulse of an amplitude 15(v) and a duration 1(μs) is applied through a series resistance of 25(Ω) to the input terminals of a 50(Ω) lossless coaxial transmission line. The line is 400(m) long and is short circuited at the far end. Determine the voltage response at the mid- point of the line as a function of time up to 8(μs). The dielectric constant of the insulating material in the cable is 2.25 10

TABLE A.3
Electrical characteristics of bare aluminum conductors steel-reinforced (ACSR)*

					Resistance			Reactance per conductor 1-ft spacing, 60 Hz	
						Ac, 60 Hz			
Code word	Aluminum area, cmil	Stranding Al/St	Layers of aluminum	Outside diameter, in	Dc, 20°C, Ω/1,000ft	20°C, Ω/mi	50°C, Ω/mi	GMR D_S ft	Inductive X_a, Ω/mi	Capacitive X'_a, MΩ.mi
Waxwing	266.800	18/1	2	0,609	0,0646	0,3488	0,3831	0,0198	0,476	0,1090
Partridge	266.800	26/7	2	0,642	0,0640	0,3452	0,3792	0,0217	0,465	0,1074
Ostrich	300.000	26/7	2	0,680	0,0569	0,3070	0,3372	0,0229	0,458	0,1057
Merlin	336.400	18/1	2	0,684	0,0512	0,2767	0,3037	0,0222	0,462	0,1055
Linnet	336.400	26/7	2	0,721	0,0507	0,2737	0,3006	0,0243	0,451	0,1040
Oriole	336.400	30/7	2	0,741	0,0504	0,2719	0,2987	0,0255	0,445	0,1032
Chickadee	397.500	18/1	2	0,743	0,0433	0,2342	0,2572	0,0241	0,452	0,1031
Ibis	397.500	26/7	2	0,783	0,0430	0,2323	0,2551	0,0264	0,441	0,1015
Pelican	477.000	18/1	2	0,814	0,0361	0,1957	0,2148	0,0264	0,441	0,1004
Flicker	477.000	24/7	2	0,846	0,0359	0,1943	0,2134	0,0284	0,432	0,0992
Hawk	477.000	26/7	2	0,858	0,0357	0,1931	0,2120	0,0289	0,430	0,0988
Hen	477.000	30/7	2	0,883	0,0355	0,1919	0,2107	0,0304	0,424	0,0980
Osprey	556.500	18/1	2	0,879	0,0309	0,1679	0,1843	0,0284	0,432	0,0981
Parakeet	556.500	24/7	2	0,914	0,0308	0,1669	0,1832	0,0306	0,423	0,0969
Dove	556.500	26/7	2	0,927	0,0307	0,1663	0,1826	0,0314	0,420	0,0965
Rook	636.000	24/7	2	0,977	0,0269	0,1461	0,1603	0,0327	0,415	0,0950
Grosbeak	636.000	26/7	2	0,990	0,0268	0,1454	0,1596	0,0335	0,412	0,0946
Drake	795.000	26/7	2	1,108	0,0215	0,1172	0,1284	0,0373	0,399	0,0912
Tern	795.000	45/7	3	1,063	0,0217	0,1188	0,1302	0,0352	0,406	0,0925
Rail	954.000	45/7	3	1,165	0,0181	0,0997	0,1092	0,0386	0,395	0,0897
Cardinal	954.000	54/7	3	1,196	0,0180	0,0988	0,1082	0,0402	0,390	0,0800
Ortolan	1.033.500	45/7	3	1,213	0,0167	0,0924	0,1011	0,0402	0,390	0,0885
Bluejay	1.113.000	45/7	3	1,259	0,0155	0,0861	0,0941	0,0415	0,386	0,0874
Finch	1.113.000	54/19	3	1,293	0,0155	0,0856	0,0937	0,0436	0,380	0,0866
Bittern	1.272.000	45/7	3	1,345	0,0136	0,0762	0,0832	0,0444	0,378	0,0855
Pheasant	1.272.000	54/19	3	1,382	0,0135	0,0751	0,0821	0,0466	0,372	0,0847
Bobolink	1.431.000	45/7	3	1,427	0,0121	0,0684	0,0746	0,0470	0,371	0,0837
Plover	1.431.000	54/19	3	1,465	0,0120	0,0673	0,0735	0,0494	0,365	0,0829
Lapwing	1.590.000	45/7	3	1,502	0,0109	0,0623	0,0678	0,0498	0,364	0,0822
Falcon	1.590.000	54/19	3	1,545	0,0108	0,0612	0,0667	0,0523	0,358	0,0814
Bluebird	2.156.000	84/19	4	1,762	0,0080	0,0476	0,0515	0,0586	0,344	0,0776

*ost used multilayer sizes.

, by permission, from Aluminum Association, *Aluminum Electrical Conductor Handbook*, 2nd ed., Washington, D.C., 1982.

W.D. Stevenson, *Power System Analysis*, McGraw-Hill, 1994

TABLE A.4 Inductive reactance spacing factor X_d at 60 Hz* (ohms per mile per conductor)

	Separation											
	Inches											
Feet	0	1	2	3	4	5	6	7	8	9	10	11
0	.....	-0,3015	-0,2174	-0,1682	-0,1333	-0,1062	-0,0841	-0,0654	-0,0492	-0,0349	-0,0221	-0,0106
1	0,0000	0,0097	0,0187	0,0271	0,0349	0,0423	0,0492	0,0558	0,0620	0,0679	0,0735	0,0789
2	0,0841	0,0891	0,0938	0,0984	0,1028	0,1071	0,1112	0,1152	0,1190	0,1227	0,1264	0,1299
3	0,1333	0,1366	0,1399	0,1430	0,1461	0,1491	0,1520	0,1549	0,1577	0,1604	0,1631	0,1657
4	0,1682	0,1707	0,1732	0,1756	0,1779	0,1802	0,1825	0,1847	0,1869	0,1891	0,1912	0,1933
5	0,1953	0,1973	0,1993	0,2012	0,2031	0,2050	0,2069	0,2087	0,2105	0,2123	0,2140	0,2157
6	0,2174	0,2191	0,2207	0,2224	0,2240	0,2256	0,2271	0,2287	0,2302	0,2317	0,2332	0,2347
7	0,2361	0,2376	0,2390	0,2404	0,2418	0,2431	0,2445	0,2458	0,2472	0,2485	0,2498	0,2511
8	0,2523											
9	0,2666											
10	0,2794											
11	0,2910											
12	0,3015											
13	0,3112											
14	0,3202											
15	0,3286											
16	0,3364											
17	0,3438											
18	0,3507											
19	0,3573											
20	0,3635											
21	0,3694											
22	0,3751											
23	0,3805											
24	0,3856											
25	0,3906											
26	0,3953											
27	0,3999											
28	0,4043											
29	0,4086											
30	0,4127											
31	0,4167											
32	0,4205											
33	0,4243											
34	0,4279											
35	0,4314											
36	0,4348											

At 60 Hz, in Ω/mi per conductor

$X_d = 0.2794 \log d$

d= separation, ft

For three-phase lines

$d = D_{eq}$

Copiado de: J.J.Grainger, W.D.Stevenson, Power System Analysis, McGraw-Hill, 1994.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

3rd Year 2nd Semester Final Examination-2022

Course Code: EEE 337 Session: 2019-20

Course Title: Power System II

Total Marks: 100 **Credits :3.00** **Time: 3 Hours**

Part A

(Answer any **two** questions)

1. (a) What is transposition and why is transposition used in transmission Lines? [2]
(b) Prove that the inductance of each phase of three-phase lines with equilateral spacing is equal. [5]
(c) Derive the expression for the flux linkages of one conductor in a group. [8]
(d) A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1(d). [10]
Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current.

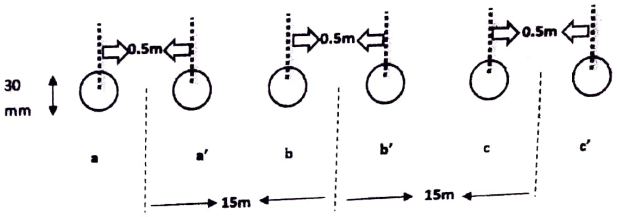


Figure: 1(d)

2. (a) What is bundled conductor & why bundled conductor is needed? Each conductor of the bundled conductor line shown in figure 2(a) is ACSR, 1232000-cmi Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. Also find the per-unit series reactance line if it's length is 160 Km base is 100 MV, 345KV. [10]

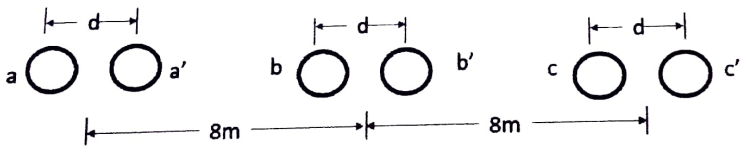


Figure 2(a)

(b) A three-phase 50Hz Transmission line has flat horizontal spacing with 3.5m between adjacent conductors. The conductors are No.2/0 hard-drawn seven-strand copper (outside conductor diameter=1.05cm). The voltage of the line is 110KV. Find the capacitance of neutral and the charging current per Km of the line. [5]
(c)Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. [10]

3. (a) What is corona? Write down the advantages and disadvantages of corona. [4]
(b) Define Underground cable. Write down the necessary requirements of underground cable. [7]
(c) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2·5 cm and the resistivity of insulation is $4{\cdot}5 \times 10^{14}$ Ω-cm, find the insulation thickness. [7]
(d) Find out the most economical conductor size in a cable. [7]

Part B

(Answer any **two** questions)

4. (a) Define (i) Thermal Resistance. (ii)Sag [4]
(b) What is critical disruptive voltage? Write down its significance. [5]
(c) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2·25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2·75:1. Calculate the distance of the fault from the testing end of the cable. [7]
(d) Prove that Dielectric stress is maximum at the conductor surface and its values go on decreasing as we move away from the conductor. [9]

5. (a) What are the main components of the overhead line? [5]
(b) What is String Efficiency? Discuss the methods of improving String Efficiency. [10]
(c) The towers of height 30 m and 90 m respectively support a transmission line conductor at the water crossing. The horizontal distance between the towers is 500 m. If the tension in the conductor is 1600 kg, find the minimum clearance of the conductor and water and clearance mid-way between the supports. The weight of the conductor is 1·5 kg/m. The bases of the towers can be considered to be at water level. [10]

8.23

6. (a) Define HVDC. Discuss the Economics of Power transmission. [5]
(b) Derive the Swing Equation for a synchronous machine. [10]
(c) The single-line diagram of figure 6(c) through parallel transmission lines to a large metropolitan system is considered as an infinite bus. The machine delivers 1.0 per unit power and both the terminal voltage and the infinite bus voltage are 1.0 per-unit. Numbers on the diagram indicate the values of the reactances on a common system base. The transient reactance of the generator is 0.2 per unit as indicated. Determine the power-angle equation for the given system operating conditions. [10]

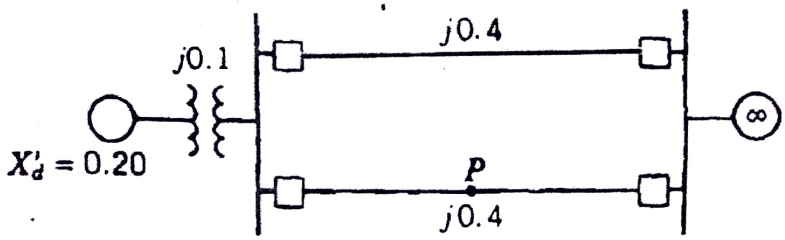


Figure 6(c)

Department of Electrical & Electronic Engineering, SUST

Course Code: EEE 337 Course Title: Power System II

Term Test 3, Total Marks: 10, Time: 20 min

1. The generator of figure 1 is delivering 1.0 pu power to the infinite bus ($|V| = 1.0\ pu$), with the generator terminal voltage ($|Vt| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. [5]

21 Final

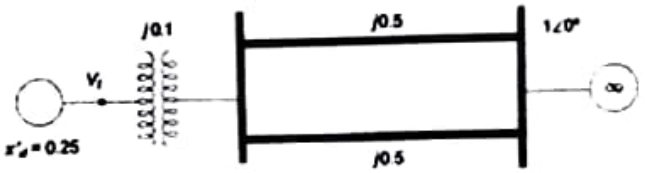


Figure: 1

2. A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. [5]

21 TT

PS TT-2

1.Varely loop test math (which answer is 15km that math)

2. Sag math (final er ta)

PS TT-1

1. Telephone line math → all-time favourite

2. Unsymmetrical spacing proof inductance/ capacitance cannot remember

Shahjalal University of Science and Technology
Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2024
Course Code: EEE 337, Session: 2020-2021
Course Title: Power System II

Full Marks: 100 Credit: 3.0 Exam Duration: 3 Hours

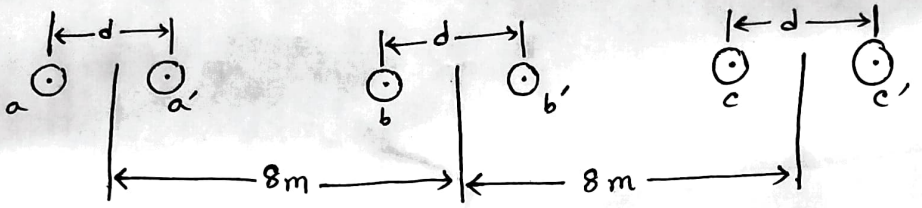
Answer any two questions from each part

Part A

1. **(a)** Define Inductance with the required expression. **[1]**

(b) Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. **[4]**

Stevenson 4.5



21-TT, Final Fig.1

(c) Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. **[10]**

(d) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. **[10]**

21 TT

2. **(a)** Define Corona Effect. **[1]**

(b) Define Transposed Transmission line. Explain the reason behind Transpose. **[4]**

(c) Derive the expression of Inductance of composite conductor lines. Ant 2.7 **[10]**

(d) A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. **[10]**

11.7

3. **(a)** Define Skin Effect. **[1]**

(b) Describe different FACTS devices and applications. **[4]**

(c) Derive the expression of dielectric stress in a single core cable. **[10]**

(d) The capacitances of a 3-phase belted cable are 12·6 μF between the three cores bunched together and the lead sheath and 7·4 μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. **[10]**

11.18

Part B

4. **(a)** Define critical disruptive voltage. Write down its significance. **[5]**

(b) What is String Efficiency? Discuss the methods of improving String Efficiency. [10]

(c) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage [10]

11.8

5. (a) What are the causes of insulation failure? [5]

(b) Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. [10]

11.24

(c) A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. [10]

8.17

6. (a) Define Power Angle. Briefly explain Equal Area Criterion. [5]

(b) Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram. [10]

(c) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. [10]

21 Final

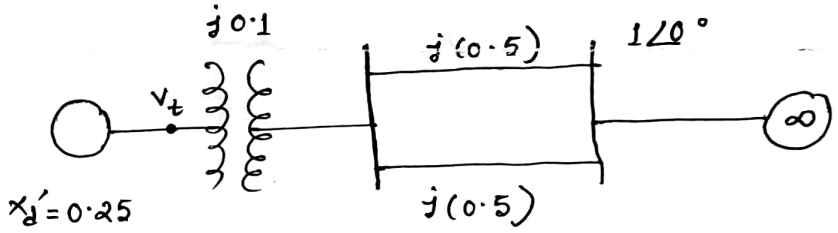


Fig. 2